Image Super-Resolution With Parametric Sparse Model Learning

Yongbo Li, Weisheng Dong, *Member, IEEE*, Xuemei Xie, Guangming Shi, *Senior Member, IEEE*, Jinjian Wu, and Xin Li, *Fellow, IEEE*

***Abstract*— Recovering a high-resolution (HR) image from its low-resolution (LR) version is an ill-posed inverse problem. Learning accurate prior of HR images is of great importance to solve this inverse problem. Existing super-resolution (SR) methods either learn a non-parametric image prior from training data (a large set of LR/HR patch pairs) or estimate a parametric prior from the LR image analytically. Both methods have their limitations: the former lacks flexibility when dealing with different SR settings; while the latter often fails to adapt to spatially varying image structures. In this paper, we propose to take a hybrid approach toward image SR by combining those two lines of ideas-that is, a parametric sparse prior of HR images is learned from the training set as well as the input LR image. By exploiting the strengths of both worlds, we can more accurately recover the sparse codes and therefore HR image patches than conventional sparse coding approaches. Experimental results show that the proposed hybrid SR method significantly outperforms existing model-based SR methods and is highly competitive to current state-of-the-art learning-based SR methods in terms of both subjective and objective image qualities.**

***Index Terms*— Image super-resolution, parametric model learning, sparse representation, deep neural networks.**

I. Introduction

In many image processing applications (e.g., high-definition televisions [1], smartphone cameras [2], video surveillance [3], and medical imaging [4], [5]), it is desirable to obtain a higher resolution (HR) image from its low-resolution (LR) observation. As fine details are lost during the acquisition, it is often challenging to recover the missing details while reconstructing the HR image from a single LR image [6], [7]. To meet this challenge, strong prior knowledge about the unknown HR image is required. In the past decades, extensive efforts have been made to design or learn an image prior for single-image super-resolution (SR) [8]–[14]. Existing image SR methods can be classified into two categories based on the way image prior is obtained: *model-based* and *learning-based*.

In model-based SR methods, single-image SR is often formulated as a Maximum a Posterior (MAP) estimation problem. When the image model incorporating the prior of HR images is served as a regularizer, HR image can be recovered by solving a regularized optimization problem. Various image priors - ranging from piecewise smoothness to sparsity and structured sparsity constraints - have been proposed for image SR [12], [13], [15]–[19]. The smoothness prior such as Tikhonov and total variation (TV) models [15] are effective for recovering piecewise smooth structures but tend to over-smooth image details. The sparsity prior, assuming that HR patches admit sparse representation with respect to a dictionary, has shown better results than smoothness prior. When combined with dictionary learning (DL), sparse models have found successful applications in image SR [12]–[14], [16]. In addition to DL, designing an appropriate sparse regularizer is critical for the success of these methods. In the past decade, parametric sparse models, e.g., zero-mean Laplacian and Generalized Gaussian models corresponding to the ℓ_1 and ℓ_p $(0 \le p \le 1)$ regularizers, have been widely adopted. Moreover, it has been shown that the sparsity-based SR performance can be further boosted by exploiting the structural self-similarity of natural images [13], [14], [16]. Though impressive SR results have been achieved by sparsity-based methods, it is still challenging to recovery HR images for large scaling factors (e.g., > 4).

In learning-based SR methods, image prior is directly learned from training data (a large set of LR/HR image patch pairs) [20]–[30], [30]–[33]. Specifically, a nonlinear mapping function between the LR and the high-frequency details of HR patches is learned. Popular approaches include sparse coding based methods [20], [34] where LR/HR dictionary pair is jointly learned from a training set, anchored neighborhood points (ANR) [21] and its extended versions (i.e., A+ [22], [27]). In ANR-based methods [21], [22], [27], training image patches are first divided into clusters, then linear mapping functions between LR and HR patches are learned for each cluster via ridge regression. Recently, deep neural networks (DNN) have also been proposed to learn the mapping functions between LR/HR patches [25], [28], [29], [33] and shown state-of-the-art SR performance [25], [28], [29].

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Y. Li and W. Dong are with the State Key Laboratory on Integrated Services Networks, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: wsdong@mail.xidian.edu.cn).

X. Xie, G. Shi, and J. Wu are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China.

X. Li is with the Lane Department of Computer Science and Electrical Engineering, West Virginia University, Morgantown, WV 26506-6109 USA.

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However, it has also been found that learning-based SR methods lack flexibilities in adapting to different SR settings (e.g., when degradation process varies).

In this paper, we propose to take a novel hybrid approach toward image SR by combining model-based and learning-based approaches. More specifically, a parametric sparse prior of HR images is learned from the training set (external source) as well as the input LR image (internal source). By exploiting the information from both sources, we can more accurately recover the sparse codes and therefore HR image patches than conventional sparse coding approaches. Experimental results show that the proposed hybrid SR method significantly outperforms existing model-based SR methods and is highly competitive to current state-of-the-art learning-based SR methods in terms of both subjective and objective image qualities.

II. Learning Parametric Sparse Models for Super-Resolution

We first present the sparse representation method using Laplacian scale mixture models and then introduce the expectation learning method for sparse coefficients.

A. Sparse Representation With Laplacian Scale Mixture Models

Let $\boldsymbol{y} = \mathbf{H}\boldsymbol{x} + \boldsymbol{n}$, $\boldsymbol{y} \in \mathbb{R}^M$ denote an observed LR image, where $\mathbf{H} \in \mathbb{R}^{M\times N}$ is the sub-sampling matrix, $\boldsymbol{x} \in \mathbb{R}^N$ and $\boldsymbol{n} \in \mathbb{R}^M$ correspond to the original image and additive Gaussian noise respectively. Assuming that each patch of the unknown image $\boldsymbol{x}$ has a sparse representation with respect to a dictionary, the sparsity-based SR can be formulated as

$$(\hat{\boldsymbol{x}}, \hat{\boldsymbol{\alpha}}_i) = \underset{\boldsymbol{x},\boldsymbol{\alpha}_i}{\operatorname{argmin}} ||\boldsymbol{y} - \mathbf{H}\boldsymbol{x}||_2^2 + \eta \sum_i \{||\mathbf{R}_i\boldsymbol{x} - \mathbf{D}\boldsymbol{\alpha}_i||_2^2 + \lambda\psi(\boldsymbol{\alpha}_i)\}, \quad (1)$$

where $\mathbf{R}_i \in \mathbb{R}^{n\times N}$ represents the matrix extracting image patch of size $\sqrt{n}\times\sqrt{n}$ at position i from $\boldsymbol{x}$, $\mathbf{D} \in \mathbb{R}^{n\times K}$ the dictionary, and $\psi(\cdot)$ the sparsity regularization term. We note that Eq. (1) is a generalization of the well-known Sparse Land model proposed in [35] for image denoising, which can be solved via alternative optimization. For a fixed estimate of $\boldsymbol{x}$, each patch is sparsely coded with $\mathbf{D}$ to obtain the sparse codes $\boldsymbol{\alpha}_i$, and then $\boldsymbol{x}$ is updated based on $\boldsymbol{\alpha}_i$ by solving the quadratic $\boldsymbol{x}$-subproblem.

Let $\boldsymbol{x}_i$ denote a patch of size $\sqrt{n} \times \sqrt{n}$. The sparse representation of $\boldsymbol{x}_i$ with respective to dictionary $\mathbf{D} \in \mathbb{R}^{n\times K}$ can be obtained by solving an ℓ_1-minimization problem

$$\hat{\boldsymbol{\alpha}}_i = \underset{\boldsymbol{\alpha}_i}{\operatorname{argmin}} ||\boldsymbol{x}_i - \mathbf{D}\boldsymbol{\alpha}_i||_2^2 + \lambda||\boldsymbol{\alpha}_i||_1. \quad (2)$$

Minimizing Eq. (2) equals to maximizing the posterior $P(\boldsymbol{\alpha}_i|\boldsymbol{x}_i)$, where the coefficients are assumed to be independent and observe Laplacian priors $P(\boldsymbol{\alpha}_i) = \prod_j (1/\theta) \exp(-|\alpha_{i,j}|/\theta)$, and λ can be computed as $\lambda = 2\sigma_n^2/(\theta+\epsilon)$, where σ_n^2 is the variance of Gaussian noise, θ is the standard deviation of $\alpha_{i,j}$, and ϵ is a small constant introduced for numerical stability. Note that if $\alpha_{i,j}$ are characterized by Laplacian distributions with different parameters $\theta_{i,j}$, then maximizing the posterior $P(\boldsymbol{\alpha}_i|\boldsymbol{x}_i)$ will lead to a weighted ℓ_1 sparsity regularization term - i.e., $\sum_j \lambda_{i,j}|\alpha_{i,j}|$, where $\lambda_{i,j}$ can be computed as $\lambda_{i,j} = 2\sigma_n^2/(\theta_{i,j}+\epsilon)$. Despite the popularity of the ℓ_1 sparse coding, the performance of the ℓ_1 sparse coding is rather limited. This is due to the inaccurate Laplacian prior, where $\alpha_{i,j}$ is assumed to be zero-mean. However, as shown in Fig. 2, when conditioned on a group of similar patches, the local means of $\boldsymbol{\alpha}_i$ can be estimated, which are often nonzero. Also, it is challenging to estimate $\theta_{i,j}$, as $\alpha_{i,j}$ is also unknown.

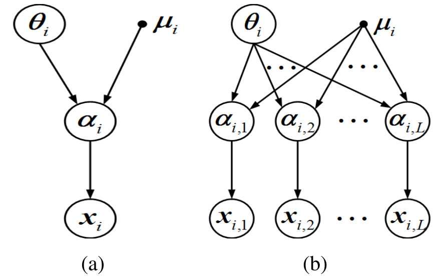


Fig. 1. Graphical model representation of the proposed LSM models, where circled nodes correspond to random variables and other nodes are model parameters. (a) The LSM model assuming each coefficient $\boldsymbol{\alpha}_i$ is independent. (b) Structured LSM model assuming a group of similar sparse codes share the same parameters of $\boldsymbol{\theta}_i$ and $\boldsymbol{\mu}_i$.

In this paper, similar to scale mixture models [16], [36], [37], we propose to model sparse coefficients by a nonzero-mean Laplacian scale mixture model (LSM). The basic idea of LSM is to impose a hyper prior on the standard deviation $\theta_{i,j}$. Then the LSM prior can be expressed by

$$P(\boldsymbol{\alpha}_i) = \prod_j P(\alpha_{i,j}),$$
$$P(\alpha_{i,j}) = \int_0^\infty P(\alpha_{i,j}|\theta_{i,j})P(\theta_{i,j})d\theta_{i,j}, \quad (3)$$

where $P(\alpha_{i,j}|\theta_{i,j})$ is a Laplacian distribution assumed to have nonzero mean and $P(\theta_{i,j})$ is the prior distribution of $\theta_{i,j}$. To facilitate the understanding of our LSM model, Fig. 1(a) shows a graphical model corresponding to the LSM model for sparse representation.

For an input patch $\boldsymbol{x}_i$, we estimate both $\boldsymbol{\alpha}_i$ and $\boldsymbol{\theta}_i$ by maximizing the posterior $P(\boldsymbol{\alpha}_i, \boldsymbol{\theta}_i|\boldsymbol{x}_i)$ as

$$\max_{\boldsymbol{\alpha}_i,\boldsymbol{\theta}_i} P(\boldsymbol{\alpha}_i, \boldsymbol{\theta}_i|\boldsymbol{x}_i) = \max_{\boldsymbol{\alpha}_i,\boldsymbol{\theta}_i}\{P(\boldsymbol{x}_i|\boldsymbol{\alpha}_i)P(\boldsymbol{\alpha}_i|\boldsymbol{\theta}_i)P(\boldsymbol{\theta}_i)\}. \quad (4)$$

where $P(\boldsymbol{x}_i|\boldsymbol{\alpha}_i)$ is the Gaussian term. Here, we have used the noninformative Jeffrey prior [38] for $P(\boldsymbol{\theta}_i)$ - i.e., $P(\theta_{i,j}) = \frac{1}{(\theta_{i,j}+\epsilon)}$. It follows that the MAP estimator can be given by

$$(\hat{\boldsymbol{\alpha}}_i, \hat{\boldsymbol{\theta}}_i) = \underset{\boldsymbol{\alpha}_i,\boldsymbol{\theta}_i}{\operatorname{argmin}} ||\boldsymbol{x}_i - \mathbf{D}\boldsymbol{\alpha}_i||_2^2 + 2\sigma_n^2 \sum_j \frac{|\alpha_{i,j} - \mu_{i,j}|}{(\theta_{i,j}+\epsilon)} + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon}), \quad (5)$$

where $\mu_{i,j}$ denotes the expectation of $\alpha_{i,j}$ (the estimation of $\boldsymbol{\mu}_i$ will be elaborated later). From Eq.(5), one can see that $\boldsymbol{\alpha}_i - \boldsymbol{\mu}_i$

are enforced to be sparse - i. e., sparse coding coefficients should be close to their means. Note that when $\boldsymbol{\mu}_i = 0$ the biased-mean sparsity regularization $||\boldsymbol{\alpha}_i - \boldsymbol{\mu}_i||_1$ reduces to the conventional ℓ_1 sparsity regularizer $||\boldsymbol{\alpha}_i||_1$. Here, we do not explicitly impose sparsity constraint on $\boldsymbol{\alpha}_i$ itself but adaptively select a compact dictionary (e.g., principle component analysis, PCA dictionary) from a set of pre-trained K PCA sub-dictionaries. This implies that sparse coding coefficients over other sub-dictionaries tend to be zero, leading to a natural sparse representation of the given patch. Meanwhile, we note that in Eq. (5) a weighted ℓ_1 sparse regularizer is derived, where the weights $1/(\boldsymbol{\theta}_i + \epsilon)$ are jointly estimated with $\boldsymbol{\alpha}_i$ by solving a minimization problem. This is in sharp contrast to existing weighted ℓ_1 sparse model where the weights are computed heuristically. In scale mixture modeling [39], $\alpha_{i,j}$ is represented by the product of a random variable $\beta_{i,j}$ and a hidden scalar multiplier $\theta_{i,j}$- i.e., $\alpha_{i,j} = \beta_{i,j}\theta_{i,j}$. Therefore Eq. (5) can be re-written into

$$(\hat{\boldsymbol{\beta}}_i, \hat{\boldsymbol{\theta}}_i) = \underset{\boldsymbol{\beta}_i, \boldsymbol{\theta}_i}{\operatorname{argmin}} ||\boldsymbol{x}_i - \mathbf{D}\Lambda_i \boldsymbol{\beta}_i||_2^2 + 2\sigma_n^2 ||\boldsymbol{\beta}_i - \boldsymbol{\gamma}_i||_1 + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon}), \quad (6)$$

where $\Lambda_i = \operatorname{diag}(\theta_{i,j}) \in \mathbb{R}^{K \times K}$ is a diagonal matrix and $\gamma_{i,j} = \mu_{i,j}/(\theta_{i,j} + \epsilon)$. From Eq. (6), one can see that the estimation of $\boldsymbol{\alpha}_i$ boils down to the joint estimation of $\boldsymbol{\beta}_i$ and $\boldsymbol{\theta}_i$ from $\boldsymbol{x}_i$.

It has been known that natural images contain abundant nonlocal repetitive structures, which can be exploited to improve the performance of sparse estimation. Let $\mathbf{X}_i = [\boldsymbol{x}_{i,1}, \boldsymbol{x}_{i,2}, \cdots, \boldsymbol{x}_{i,L}]$ denote a set of patches similar to exemplar patch $\boldsymbol{x}_i$ (including $\boldsymbol{x}_i$ itself) and $\mathbf{A}_i = [\boldsymbol{\alpha}_{i,1}, \boldsymbol{\alpha}_{i,2}, \cdots, \boldsymbol{\alpha}_{i,L}]$ be the corresponding sparse codes of $\mathbf{X}_i$. The structured correlations among $\mathbf{A}_i$ can be well characterized with LSM model by imposing the same prior (i.e., identical $\boldsymbol{\mu}_i$ and $\boldsymbol{\theta}_i$) on similar patches. The prior of $\mathbf{A}_i$ can then be expressed by

$$P(\mathbf{A}_i) = \prod_l \prod_j P(\alpha_{l,j}),$$
$$P(\alpha_{l,j}) = \int_0^\infty P(\alpha_{l,j}|\theta_{i,j}) P(\theta_{i,j}) d\theta_{i,j}. \quad (7)$$

The corresponding graphical model is shown in Fig. 1 (b). By substituting $P(\mathbf{A}_i)$ into the MAP estimator, we obtain the following objective function for joint estimation of $\mathbf{A}_i$,

$$(\hat{\mathbf{B}}_i, \hat{\boldsymbol{\theta}}_i) = \underset{\mathbf{B}_i, \boldsymbol{\theta}_i}{\operatorname{argmin}} ||\mathbf{X}_i - \mathbf{D}\Lambda_i \mathbf{B}_i||_F^2 + 2\sigma_n^2 \sum_{l=1}^{L} ||\boldsymbol{\beta}_l - \boldsymbol{\gamma}_i||_1 + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon}), \quad (8)$$

where $\mathbf{A}_i = \Lambda_i \mathbf{B}_i$ and $\mathbf{B}_i = [\boldsymbol{\beta}_{i,1}, \boldsymbol{\beta}_{i,2}, \cdots, \boldsymbol{\beta}_{i,L}]$. In [36] and [37], group LSM model with Gamma distribution has also been proposed for compressive sensing applications. However, the variance parameter derived in the LSM of [36] is similar to the weights derived in reweighted ℓ_1-norm regularizer. More general scale mixture models for group-sparse modeling have also been proposed in [37], where variational Bayesian inference approach was used to derive an estimation procedure. The proposed group LSM model differs from these works in that the variance parameters are jointly optimized along with sparse coding coefficients from the observed data. As shown in the next section, efficient optimization algorithm for solving the objective function of Eq. (8) can be derived.

B. Image Super-Resolution Based on the Parametric Sparse Models

Based on the structured sparse model of Eq. (8), the proposed image SR method can be formulated as

$$(\hat{\boldsymbol{x}}, \hat{\mathbf{B}}_i, \hat{\boldsymbol{\theta}}_i) = \underset{\boldsymbol{x}, \mathbf{B}_i, \boldsymbol{\theta}_i}{\operatorname{argmin}} ||\boldsymbol{y} - \mathbf{H}\boldsymbol{x}||_2^2 + \eta \sum_i \{||\tilde{\mathbf{R}}_i \boldsymbol{x} - \mathbf{D}\Lambda_i \mathbf{B}_i||_F^2 + 2\sigma_n^2 ||\mathbf{B}_i - \boldsymbol{\Gamma}_i||_1 + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon})\}, \quad (9)$$

where $\tilde{\mathbf{R}}_i \boldsymbol{x} \doteq [\mathbf{R}_{i,1}\boldsymbol{x}, \mathbf{R}_{i,2}\boldsymbol{x}, \cdots, \mathbf{R}_{i,L}\boldsymbol{x}] \in \mathbb{R}^{n \times L}$ denotes the matrix formed by the set of patches similar to $\boldsymbol{x}_i$ (including $\boldsymbol{x}_i$ itself), $\mathbf{R}_i$ denotes the matrix extracting the patch $\boldsymbol{x}_i$ at position i, and $\boldsymbol{\Gamma}_i = [\boldsymbol{\gamma}_i, \cdots, \boldsymbol{\gamma}_i] \in \mathbb{R}^{K \times L}$. In Eq. (9), we use the structured sparse model to regularize the reconstructed HR image $\boldsymbol{x}$- i.e., the group of similar patches $\mathbf{X}_i = \tilde{\mathbf{R}}_i \boldsymbol{x}$ extracted from $\boldsymbol{x}$ should have sparse representation with respective to $\mathbf{D}$ and the distributions of sparse coefficients are characterized by the same LSM prior.

In sparsity-based SR methods, the selection of the dictionary often has a direct impact on the performance. Instead of learning an overcomplete dictionary $\mathbf{D}$, which has high computational complexity, we propose to learn a set of orthogonal bases for each local texture and edge patterns. Similar to [17], the set of training image patches are first clustered into clusters; then a PCA basis $\mathbf{D}_k$ is computed for each cluster to compactly represent the patches associated with this cluster. For each exemplar patch $\boldsymbol{x}_i$, we first cluster $\boldsymbol{x}_i$ into a cluster (denoted as the k_i-th cluster) by a clustering method (e.g., K-means), and then the corresponding PCA basis $\mathbf{D}_{k_i}$ is used to code $\boldsymbol{x}_i$ as well as those similar to $\boldsymbol{x}_i$. Additionally, the expectations $\boldsymbol{\mu}_i = \Lambda_i \boldsymbol{\gamma}_i$ of sparse codes $\boldsymbol{\alpha}_i$ in Eq. (9) are important to the performance of the proposed SR method. Therefore in the next subsection, we will present a novel method of learning $\boldsymbol{\mu}_i$ from both external training HR images and internal similar patches of the input image.

C. Learning Expectations $\boldsymbol{\mu}$ of the Sparse Codes

We aim at learning the expectation $\boldsymbol{\mu}_i$ of $\boldsymbol{\alpha}_i$ for each exemplar HR patch $\boldsymbol{x}_i$. Since we only have access to LR image $\boldsymbol{y}$, there are generally two ways of estimating $\boldsymbol{\mu}_i$. First, we can use a sparsity-based SR method to reconstruct a HR image, denoted as $\hat{\boldsymbol{x}}$, from which sparse codes $\boldsymbol{\alpha}_i$ can be estimated. Using the estimated $\hat{\boldsymbol{\alpha}}_i$, the expectation $\boldsymbol{\mu}_i$ can then be estimated as a weighted average $\hat{\boldsymbol{\mu}}_i = \sum_l w_l \hat{\boldsymbol{\alpha}}_{i,l}$, where $\hat{\boldsymbol{\alpha}}_{i,l}$ are the sparse codes corresponding to image patches $\hat{\boldsymbol{x}}_{i,l}$ similar to $\hat{\boldsymbol{x}}_i$. In [13], such nonlocal estimation of the sparse prior has led to excellent sparse estimation performance. However, the effectiveness of such nonlocal estimation degrades rapidly in the situation of large scaling factors (e.g., larger than 4).

An alternative approach of estimating $\boldsymbol{\mu}_i$ is to leverage the information from external training HR images. The basic idea is to learn a mapping function that maps an input LR patch $\boldsymbol{y}_i \in \mathbb{R}^m$ to an estimate of $\boldsymbol{\alpha}_i$. Toward this objective, we can define the following learning function without loss of generality

$$\tilde{\boldsymbol{\alpha}}_i = f(z_i; \mathbf{W}, \boldsymbol{b}) = g(\mathbf{W} * z_i + \boldsymbol{b}), \tag{10}$$

where z_i denotes the feature vector extracted from $\boldsymbol{y}_i$, $\mathbf{W}$ is the weighting matrix and $\boldsymbol{b}$ is the bias, and $g(\cdot)$ denotes a nonlinear activation function (e.g., the sigmod function). To learn the parameters ($\mathbf{W}$, $\boldsymbol{b}$), we first collect a set of LR and HR image pairs and reconstruct the LR images using a sparsity-based SR method (will describe the details later). Then the feature vectors and the corresponding HR patches are extracted from initially recovered HR images and original images respectively, forming the pairs of training patches $\{z_i, \boldsymbol{x}_i\}$, $i = 1, 2, \cdots, N$. For each $\boldsymbol{x}_i$, we obtain its sparse codes $\boldsymbol{\alpha}_i$ with respect to dictionary $\mathbf{D}$. The parameters $\mathcal{W} = \{\mathbf{W}, \boldsymbol{b}\}$ can then be optimized over the set of training patch pairs by minimizing the following objective function,

$$(\hat{\mathbf{W}}, \hat{\boldsymbol{b}}) = \underset{\mathbf{W}, \boldsymbol{b}}{\operatorname{argmin}} \sum_{i=1}^{N} ||\boldsymbol{\alpha}_i - g(\mathbf{W} * z_i + \boldsymbol{b})||_2^2, \tag{11}$$

which can be solved by using a stochastic gradient descent approach.

In this paper, instead of learning a complex nonlinear mapping function, we propose to learn a set of linear mapping functions for each different local image edge and texture patterns. Specifically, we first cluster feature vectors z_i into K clusters via standard K-means method and then learn a simple linear mapping function for each cluster. After clustering, LR/HR patches within each cluster generally contain similar structures; therefore even linear mapping functions would be sufficient for characterizing the correlations between feature vectors and sparse codes. For each cluster S_k, we propose to learn the mapping functions via minimizing

$$(\hat{\mathbf{W}}_k, \hat{\boldsymbol{b}}_k) = \underset{\mathbf{W}_k, \boldsymbol{b}_k}{\operatorname{argmin}} \sum_{i \in S_k} ||\boldsymbol{\alpha}_i - (\mathbf{W}_k z_i + \boldsymbol{b}_k)||_2^2. \tag{12}$$

For simplicity, the bias term $\boldsymbol{b}_k$ in the above equation can be absorbed into $\mathbf{W}_k$ by rewriting $\mathbf{W}_k$ and z_i as $\mathbf{W}_k = [\mathbf{W}_k, \boldsymbol{b}_k]$ and $z_i = [z_i^\top; 1]^\top$ respectively. Then the parameters $\mathbf{W}_k$ can be easily solved by standard least-square method.

After clustering the LR/HR patch pairs into K clusters, a set of compact dictionaries (i.e., the PCA bases used in Eq. (18)) can also be learned using HR patches $\boldsymbol{x}_i$. For each cluster S_k, we use the patches $\boldsymbol{x}_i$, $i \in S_k$ to compute the PCA basis. With the computed PCA basis $\mathbf{D}_k$, the sparse codes $\boldsymbol{\alpha}_i$ of $\boldsymbol{x}_i$ used in Eq. (12) can be easily calculated via soft-thresholding $\boldsymbol{\alpha}_i = \mathcal{S}(\mathbf{D}_k^\top \boldsymbol{x}_i, \lambda)$, $i \in S_k$, where $\mathcal{S}(\cdot, \lambda)$ denotes the soft-thresholding operator with a threshold of λ. The sparse codes learning algorithm is summarized in **Algorithm 1**.

With the learned mapping functions, an estimates of $\boldsymbol{\alpha}_i$ can be obtained as $\tilde{\boldsymbol{\alpha}}_i = \mathbf{W}_{k_i} z_i$, where $\mathbf{W}_{k_i}$ denotes the corresponding projection matrix of cluster k_i which the feature vector z_i belongs to. To compute the expectation $\boldsymbol{\mu}_i$ of $\boldsymbol{\alpha}_i$, we collect a set of similar patches $\hat{\boldsymbol{x}}_{i,l}$, $l = 1, 2, \cdots, L$ from the initially reconstructed HR image $\hat{\boldsymbol{x}}$, and extract feature vectors $z_{i,l}$ of the corresponding patches. Then the parameter $\boldsymbol{\mu}_i$ can be estimated by

Algorithm 1 Sparse codes learning algorithm

Initialization:

(a) Construct a set of LR and HR image pairs $\{\boldsymbol{y}, \boldsymbol{x}\}$ and recover the HR images $\{\hat{\boldsymbol{x}}\}$ with a sparsity-based SR method;
(b) Extract feature patches $\boldsymbol{z}_i$, the LR patches $\boldsymbol{y}_i$ and HR patches $\boldsymbol{x}_i$ from $\{\hat{\boldsymbol{x}}, \boldsymbol{y}, \boldsymbol{x}\}$, respectively;
(c) Clustering $\{\boldsymbol{z}_i, \boldsymbol{y}_i, \boldsymbol{x}_i\}$ into K clusters using K-means algorithm.

Outer loop: Iteration on $k = 1, 2, \cdots, K$

(a) Calculate the PCA basis $\mathbf{D}_k$ for each cluster using the HR patches belong to the k-th cluster;
(b) Computer the sparse codes as $\boldsymbol{\alpha}_i = \mathcal{S}(\mathbf{D}_{k_i}^\top \boldsymbol{x}_i, \lambda)$ for each $\boldsymbol{x}_i$, $i \in S_k$;
(c) Learn the parameters $\mathcal{W}$ of the mapping function via solving Eq. (12).
End for

Output: $\{\mathbf{D}_k, \mathcal{W}_k\}$.

$$\tilde{\boldsymbol{\mu}}_i = \sum_{l=1}^{L} w_{i,l} (\mathbf{W}_{k_i} z_{i,l}) = \mathbf{W}_{k_i} \sum_{l=1}^{L} w_{i,l} z_{i,l}, \tag{13}$$

where $w_{i,l} = \frac{1}{c} \exp(-||\hat{\boldsymbol{x}}_{i,l} - \hat{\boldsymbol{x}}_i||/h)$, c is the normalization constant and h is a predefined parameter.

Additionally, we can also obtain the expectation of $\boldsymbol{\alpha}_i$ directly from the current estimate of sparse codes $\boldsymbol{\alpha}_i$- i.e.,

$$\hat{\boldsymbol{\mu}}_i = \sum_{l=1}^{L} w_{i,l} \hat{\boldsymbol{\alpha}}_{i,l}. \tag{14}$$

where $\hat{\boldsymbol{\alpha}}_{i,l}$ denotes the estimate of $\boldsymbol{\alpha}_{i,l}$ obtained from the previous iteration. An improved estimation of $\boldsymbol{\mu}_i$ can then be obtained by combining the above two estimates - i.e.,

$$\boldsymbol{\mu}_i = \Delta \tilde{\boldsymbol{\mu}}_i + (1 - \Delta) \hat{\boldsymbol{\mu}}_i. \tag{15}$$

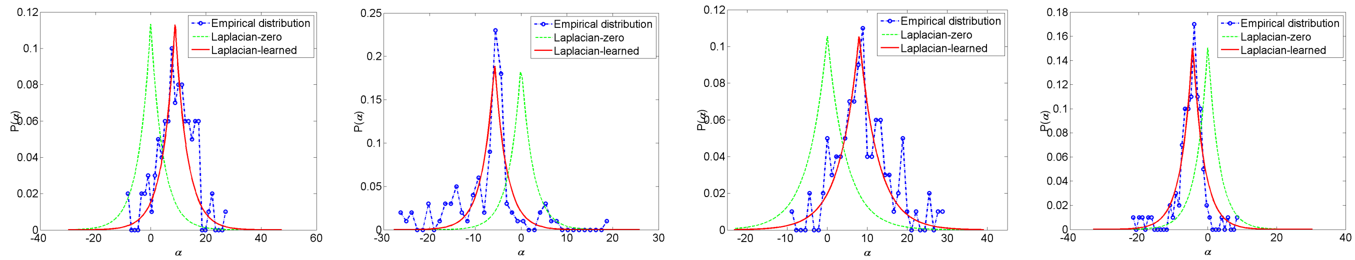
where $\Delta = \omega \text{diag}(\delta_j) \in \mathbb{R}^{K \times K}$. Similar to [14], δ_j is set according to the energy ratio of $\tilde{\mu}_{i,j}$ and $\hat{\mu}_{i,j}$ as

$$\delta_j = \frac{r_j^2}{r_j^2 + 1/r_j^2}, \quad r_j = \tilde{\mu}_{i,j} / \hat{\mu}_{i,j}. \tag{16}$$

where ω is a predefined constant.

Fig. 2 shows some empirical distributions of sparse coefficients $\boldsymbol{\alpha}_{i,l}$ for a set of similar patches collected from HR image *Butterfly*, traditional zero-mean Laplacian distributions (denoted as Laplacian-zero), and the Laplacian distributions with learned means (denoted as Laplacian-Learned). It can be seen that the local means of the coefficients $\boldsymbol{\alpha}_{i,j}$ can be nonzero and the Laplacian distributions with learned means approximate the empirical distributions much better than the zero-mean Laplacian distributions, demonstrating the effectiveness of the proposed biased-mean estimation technique.

Regarding feature vectors z_i, instead of extracting them from the LR image $\boldsymbol{y}$ directly, we extract them from an



(a) (b) (c) (d)

Fig. 2. Representation coefficients distributions for similar patches. (a)- (d) show four examples of such distributions of *Butterfly* image. The dashed dots curves show the empirical distributions of the sparse coefficients of the original HR patches, the green and red curves show the Laplacian distributions with zero mean and the learned means, respectively.

initially recovered HR image $\hat{\boldsymbol{x}}$ (e.g., obtained by applying a sparsity-based SR method). The reason is that $\hat{\boldsymbol{x}}$ contains more details than $\boldsymbol{y}$ and more accurate features can be extracted from $\hat{\boldsymbol{x}}$. In this paper, we solve the following objective function for initial estimates of HR images

$$(\hat{\boldsymbol{x}}, \hat{\mathbf{A}}_i) = \underset{\boldsymbol{x},\mathbf{A}_i}{\operatorname{argmin}}\, ||\boldsymbol{y} - \mathbf{H}\boldsymbol{x}||_2^2 + \eta \sum_i \{||\tilde{\mathbf{R}}_i \boldsymbol{x} - \mathbf{D}_{k_i}\mathbf{A}_i||_F^2 + \lambda ||\mathbf{A}_i - \hat{\mathbf{U}}_i||_1\}, \quad (17)$$

where $\mathbf{A}_i = [\boldsymbol{\alpha}_{i,1}, \boldsymbol{\alpha}_{i,2}, \cdots, \boldsymbol{\alpha}_{i,L}]$, $\mathbf{D}_{k_i}$ denotes the corresponding PCA basis assigned to example patch $\boldsymbol{x}_i$, $\hat{\mathbf{U}} = [\hat{\boldsymbol{\mu}}_i, \hat{\boldsymbol{\mu}}_i, \cdots, \hat{\boldsymbol{\mu}}_i]$, and $\hat{\boldsymbol{\mu}}_i$ denotes the nonlocal means estimation of $\boldsymbol{\alpha}_i$ using Eq. (14). The SR method of Eq. (17) is similar to the NCSR method developed in [13], except that we compute $\hat{\boldsymbol{\mu}}_i$ for a group of similar patches to save computational complexity. The optimization problem of Eq. (17) can be solved by alternative optimization, which will be detailed in the next section.

III. Optimization Algorithm for Image SR

With the learned PCA dictionaries, we can rewrite the proposed image SR problem of Eq. (9) into

$$(\hat{\boldsymbol{x}}, \hat{\mathbf{B}}_i, \hat{\boldsymbol{\theta}}_i) = \underset{\boldsymbol{x},\mathbf{B}_i,\boldsymbol{\theta}_i}{\operatorname{argmin}}\, ||\boldsymbol{y} - \mathbf{H}\boldsymbol{x}||_2^2 + \eta \sum_i \{||\tilde{\mathbf{R}}_i \boldsymbol{x} - \mathbf{D}_{k_i}\Lambda_i\mathbf{B}_i||_F^2 + 2\sigma_n^2 ||\mathbf{B}_i - \boldsymbol{\Gamma}_i||_1 + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon})\}, \quad (18)$$

where $\mathbf{D}_{k_i}$ denotes the corresponding PCA basis of cluster k_i which the exemplar patch $\boldsymbol{x}_i$ belongs to. The objective function is then approximately solved by the method of alternative optimization.

A. Solving for $\mathbf{B}_i$ and $\boldsymbol{\theta}_i$ for Fixed $\boldsymbol{x}$

For a fixed $\boldsymbol{x}$ and learned expectations $\boldsymbol{\gamma}_i$, $(\mathbf{B}_i, \boldsymbol{\theta}_i)$ can be estimated via minimizing

$$(\hat{\mathbf{B}}_i, \hat{\boldsymbol{\theta}}_i) = \underset{\mathbf{B}_i,\boldsymbol{\theta}_i}{\operatorname{argmin}}\, ||\tilde{\mathbf{R}}_i \boldsymbol{x} - \mathbf{D}_{k_i}\Lambda_i\mathbf{B}_i||_F^2 + 2\sigma_n^2 ||\mathbf{B}_i - \boldsymbol{\Gamma}_i||_1 + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon}). \quad (19)$$

For fixed $\boldsymbol{\theta}_i$, $\mathbf{B}_i$ can be solved by minimizing the following objective function,

$$\hat{\mathbf{B}}_i = \underset{\mathbf{B}_i}{\operatorname{argmin}}\, ||\tilde{\mathbf{R}}_i \boldsymbol{x} - \mathbf{D}_{k_i}\Lambda_i\mathbf{B}_i||_F^2 + 2\sigma_n^2 ||\mathbf{B}_i - \boldsymbol{\Gamma}_i||_1. \quad (20)$$

Since $\mathbf{D}_{k_i}$ is orthogonal, the above equation can be rewritten as

$$\begin{aligned}\hat{\mathbf{B}}_i &= \underset{\mathbf{B}_i}{\operatorname{argmin}}\, ||\mathbf{A}_i - \Lambda_i\mathbf{B}_i||_F^2 + 2\sigma_n^2 ||\mathbf{B}_i - \boldsymbol{\Gamma}_i||_1 \\ &= \underset{\mathbf{B}_i}{\operatorname{argmin}} \sum_{l=1}^{L} \{||\boldsymbol{\alpha}_{i,l} - \Lambda_i\boldsymbol{\beta}_{i,l}||_2^2 + 2\sigma_n^2 ||\boldsymbol{\beta}_{i,l} - \boldsymbol{\gamma}_i||_1\} \\ &= \underset{\mathbf{B}_i}{\operatorname{argmin}} \sum_{l=1}^{L} \{||\Lambda_i(\Lambda_i^{-1}\boldsymbol{\alpha}_{i,l} - \boldsymbol{\beta}_{i,l})||_2^2 + 2\sigma_n^2 ||\boldsymbol{\beta}_{i,l} - \boldsymbol{\gamma}_i||_1\},\end{aligned} \quad (21)$$

where $\mathbf{A}_i = \mathbf{D}_{k_i}^\top(\tilde{\mathbf{R}}_i \boldsymbol{x})$, $\Lambda_i = \operatorname{diag}(\theta_{i,j})$ and $\boldsymbol{\gamma}_i = \boldsymbol{\mu}_i/(\boldsymbol{\theta}_i + \epsilon)$. Therefore $\mathbf{B}_i$ can be solved in closed-form solution by

$$\begin{aligned}\hat{\mathbf{B}}_i &= \mathcal{S}(\Lambda_i^{-1}\boldsymbol{\alpha}_{i,j} - \boldsymbol{\Gamma}_i, \Lambda_i^{-1}\mathbf{T}) + \boldsymbol{\Gamma}_i \\ &= \Lambda_i^{-1}\mathcal{S}(\mathbf{D}_{k_1}^\top \tilde{\mathbf{R}}_i \boldsymbol{x} - \mathbf{U}_i, \mathbf{T}) + \boldsymbol{\Gamma}_i,\end{aligned} \quad (22)$$

where $\mathbf{T} = [\boldsymbol{t}_i, \boldsymbol{t}_i, \cdots, \boldsymbol{t}_i] \in \mathbb{R}^{K\times L}$, $\boldsymbol{t}_i = \sigma_n^2/(\boldsymbol{\theta}_i + \epsilon)$, and $\mathcal{S}(\cdot, t)$ denotes the soft-thresholding function with threshold t. For fixed $\mathbf{B}_i$, the variances $\boldsymbol{\theta}_i$ can be solved via minimizing

$$\hat{\boldsymbol{\theta}}_i = \underset{\boldsymbol{\theta}_i}{\operatorname{argmin}}\, ||\mathbf{A}_i - \Lambda_i\mathbf{B}_i||_F^2 + 4\sigma_n^2 \log(\boldsymbol{\theta}_i + \boldsymbol{\epsilon}), \quad (23)$$

where we have used $\mathbf{A}_i = \mathbf{D}_{k_i}^\top \tilde{\mathbf{R}}_i \boldsymbol{x}$. Note that the above minimization problem can be reduced into a set of scalar minimization problems - i.e.,

$$\hat{\boldsymbol{\theta}}_i = \underset{\boldsymbol{\theta}_i}{\operatorname{argmin}} \sum_j a_j\theta_{i,j}^2 + b_j\theta_{i,j} + c\log(\theta_{i,j} + \epsilon), \quad (24)$$

where $a_j = ||\boldsymbol{\beta}_i^j||_2^2$, $b_j = -2\boldsymbol{\alpha}_i^j(\boldsymbol{\beta}_i^j)^\top$, $c = 4\sigma_n^2$, $\boldsymbol{\beta}_i^j$ and $\boldsymbol{\alpha}_i^j$ denote the j-th row of $\mathbf{B}_i$ and $\mathbf{A}_i$, respectively. Each scalar minimization problem can be solved by taking $\frac{df(\theta_{i,j})}{d\theta_{i,j}} = 0$, where $f(\theta_{i,j})$ denotes the right hand side of Eq. (24). Then, we can derive the following two stationary points of $f(\theta_{i,j})$

$$\begin{aligned}\theta_{i,j}^1 &= -\frac{2a_j\epsilon + b_j}{4a_j} + \sqrt{\frac{(2a_j\epsilon + b_j)^2}{16a_j^2} - \frac{b_j + \epsilon + c}{2a_j}} \\ \theta_{i,j}^2 &= -\frac{2a_j\epsilon + b_j}{4a_j} - \sqrt{\frac{(2a_j\epsilon + b_j)^2}{16a_j^2} - \frac{b_j + \epsilon + c}{2a_j}},\end{aligned} \quad (25)$$

when $(2a_j\epsilon + b_j)^2/(16a_j^2) - (b_j\epsilon + c)/(2a_j) \geq 0$. Since ϵ is a very small positive constant, $g(\theta_{i,j}) = \frac{df(\theta_{i,j})}{\theta_{i,j}}$ is always positive at $\theta_{i,j} = 0$. Thus, $f(\theta_{i,j}^2)$ is always larger than $f(0)$.

Algorithm 2 Image SR with Learned Sparse Representation

Initialization:

(a) Initialize $\hat{\boldsymbol{x}}^{(0)}$ with a conventional SR method;
(b) Set parameters η and σ_n;

Outer loop: Iteration over $t = 0, 1, \cdots, T$

(a) Extract feature vectors $\boldsymbol{z}_i$ from $\hat{\boldsymbol{x}}^{(t)}$ and cluster the patches into clusters;
(b) Learn $\tilde{\boldsymbol{\beta}}_i$ for each local patch using Eq. (13);
(c) Update the estimate of $\boldsymbol{\beta}_i$ using Eq. (15);
(d) **Inner loop** (solve Eq.(18)): iteration over $s = 1, 2, \cdots, S$;
  (I) Compute $\mathbf{B}_i^{(s+1)}$ using Eq. (22);
  (II) Compute $\boldsymbol{\theta}_i^{(s+1)}$ using Eq. (26);
  (III) Update the whole image $\hat{\boldsymbol{x}}^{(s+1)}$ via Eq. (28);
  (III) Set $\boldsymbol{x}^{(t+1)} = \boldsymbol{x}^{(s+1)}$ if $s = S$.
  End for

Output: $\boldsymbol{x}^{(t+1)}$.

Then, the global minimizer of $f(\theta_{i,j})$ can be obtained by comparing $f(0)$ and $f(\theta_{i,j}^1)$. When $(2a_j\epsilon + b_j)^2/(16a_j^2) - (b_j\epsilon + c)/(2a_j) < 0$, there are no stationary points in the range of $[0, \infty)$. Since $g(\theta_{i,j}) = \frac{df(\theta_{i,j})}{\theta_{i,j}}$ is always positive at $\theta_{i,j} = 0$, $f(0)$ is the global minimizer for this case. In summary, the solution to Eq. (24) can be written as

$$\hat{\theta}_{i,j} = \begin{cases} 0, & \text{if } (2a_j\epsilon + b_j)^2/(16a_j^2) - (b_j\epsilon + c)/(2a_j) < 0, \\ \delta_{i,j}, & \text{otherwise} \end{cases} \tag{26}$$

where $\delta_{i,j} = \operatorname{argmin}_{\theta_{i,j}}\{f(0), f(\theta_{i,j}^1)\}$. By alternatively estimating $\boldsymbol{\theta}_j$ and $\mathbf{B}_i$, the set of similar patches can be reconstructed as $\mathbf{X}_i = \mathbf{D}_{k_i}\Lambda_i\mathbf{B}_i$.

B. Solving $\boldsymbol{x}$ for Fixed $\boldsymbol{\theta}_i$ and $\boldsymbol{B}_i$

With the estimated sparse codes, the HR image can be reconstructed by solving

$$\hat{\boldsymbol{x}} = \operatorname*{argmin}_{\boldsymbol{x}} ||\boldsymbol{y} - \mathbf{H}\boldsymbol{x}||_2^2 + \eta \sum_i ||\tilde{\mathbf{R}}_i\boldsymbol{x} - \mathbf{X}_i||_F^2 \tag{27}$$

where $\mathbf{X}_i = \mathbf{D}_{k_i}\Lambda_i\mathbf{B}_i$. The above equation admits a closed-form solution, as

$$\hat{\boldsymbol{x}} = (\mathbf{H}^\top\mathbf{H} + \eta\sum_i \tilde{\mathbf{R}}_i^\top\tilde{\mathbf{R}}_i)^{-1}(\mathbf{H}^\top\boldsymbol{y} + \eta\sum_i \tilde{\mathbf{R}}_i\mathbf{X}_i), \tag{28}$$

where $\tilde{\mathbf{R}}_i^\top\tilde{\mathbf{R}}_i = \sum_{l=1}^L \mathbf{R}_l^\top\mathbf{R}_l$ and $\tilde{\mathbf{R}}_i^\top\mathbf{X}_i = \sum_{l=1}^L \mathbf{R}_l^\top\boldsymbol{x}_{i,l}$. Since the matrix to be inverted in Eq. (28) is large, the conjugate gradient algorithm is used to compute Eq. (28). The proposed image SR method is summarized in **Algorithm 2**. In **Algorithm 2**, we iteratively extract the feature patches from an intermediately recovered image $\hat{\boldsymbol{x}}^{(t)}$ and learn $\tilde{\boldsymbol{\mu}}_i$ from the training set, leading to further improvements in estimating the expectations.

IV. Experimental Results

In this section, we report the performance of the proposed sparse model learning based SR method (denoted as SML-SR). To learn the parametric sparse models, we have used the 91 training images used in [20]. LR images are simulated by two subsampling methods, i.e., the bicubic function and the Gaussian blurring followed by subsampling. Total $100,000$ pairs of feature and HR patches of size 7×7 are extracted from the pre-recovered HR images and the original HR images, respectively. Similar to [20] and [22], the PCA is used to reduce the dimension of the feature vectors. All the feature vectors are clustered into 1000 clusters. Other major parameters of the proposed method are set as follows: $L = 12$, $T = 8$, and $J = 10$. To verify the effectiveness of the joint estimation of the standard deviations $\boldsymbol{\theta}_i$ and the sparse codes, we also implement a variant of the proposed method that uses non-weighted ℓ-norm, formulated as

$$(\hat{\boldsymbol{x}}, \hat{\mathbf{A}}_i) = \operatorname*{argmin}_{\boldsymbol{x}, \mathbf{A}_i} ||\boldsymbol{y} - \mathbf{H}\boldsymbol{x}||_2^2 + \eta\sum_i\{||\tilde{\mathbf{R}}_i\boldsymbol{x} - \mathbf{D}_{k_i}\mathbf{A}_i||_F^2 + \lambda||\mathbf{A}_i - \mathbf{U}_i||_1\}, \tag{29}$$

where the regularization parameter is manually optimized for better performance. This variant is denoted as SML-SR-Baseline. For SML-SR-Baseline, we adopt a set of parameters in a reasonable range and select the parameters which lead to the best results on the test set, i.e., Set5, Set14 and BSD100. Here we use parts of BSD100 to reduce the cost. Moreover, to verify the effectiveness of the expectation learning from external training images, we also implement another variant of the proposed method, which estimates the expectations of the sparse codes only from the nonlocal similar patches of the LR images. Let this variant be denoted as SML-SR-NLM.

We have compared the proposed SR methods with several well-known image SR methods - i.e., the sparse coding based SR method (denoted as SCSR) [20], the SR method based on sparse regression and natural prior (denoted as KK) [40], the A+ method [22], the NCSR method [13], the SR method using CNN (denoted as SRCNN [25]), and the FSRCNN method [29] that further significantly improves the performance of SRCNN in terms of both effectiveness and efficiency.[1] Note that NCSR is among the state-of-the-art sparsity-based SR methods; a list of competing methods are summarized in Table I. Three commonly used image datasets (i.e., Set5 [41], Set14 [42] and BSD100 [43]), which consist of 5, 14 and 100 images respectively, are employed as test images. Since LR images are often corrupted by noise, we also add additive Gaussian noise of standard deviation 5 to LR images, making the SR problem more challenging. All experimental results produced by the competing methods are available at our project website.[2]

A. LR Images Generated With Bicubic Interpolation Function

In this subsection, we perform image SR reconstruction from LR images that are simulated by applying the *bicubic*

[1]We thank Yu *et al.* [13], Egiazarian and Katkovnik [20], Timofte *et al.* [22], Dong *et al.* [25], Kim *et al.* [29], and Andrews and Mallows [40] for providing their codes.

[2]http://see.xidian.edu.cn/faculty/wsdong/ImageSRProject.htm

TABLE I
THE COMPETING METHODS INVOLVED IN THE COMPARISON STUDIES

Comparing Methods	Description
SCSR [21]	Sparse coding based image SR method.
KK [41]	Regularized kernel ridge regression based SR method.
A+ [23]	Improved Anchored Neighborhood Regression (ANR) method.
SRCNN [26]	Convolutional Neural Network (CNN) based SR method.
NCSR [14]	Nonlocally centralized sparse representation based SR method.
FSRCNN [30]	Advanced version of SRCNN with better SR performance.

TABLE II
AVERAGE PSNR AND SSIM RESULTS OF THE TEST METHODS (BICUBIC DOWNSAMPLING AND $\sigma_n = 0$)

Images	Se5			Set14			BSD100		
Upscaling	×2	×3	×4	×2	×3	×4	×2	×3	×4
SCSR [21]	-	31.42 0.8821	-	-	28.31 0.7954	-	-	26.54 0.7729	-
KK [41]	36.22 0.9514	32.29 0.9037	30.03 0.8544	32.12 0.9029	28.39 0.8135	27.15 0.7422	31.08 0.8834	28.15 0.7780	26.69 0.7017
A+ [23]	36.55 0.9544	32.59 0.9088	30.29 0.8603	32.28 0.9056	29.13 0.8188	27.33 0.7491	31.21 0.8863	28.29 0.7835	26.82 0.7087
SRCNN [26]	36.66 0.9542	32.75 0.9090	30.49 0.8628	32.45 0.9067	29.30 0.8215	27.50 0.7513	31.36 0.8879	28.41 0.7863	26.90 0.7103
NCSR [14]	36.68 0.9550	33.05 0.9149	30.77 0.8720	32.26 0.9058	29.30 0.8239	27.52 0.7563	31.14 0.8863	28.37 0.7872	26.91 0.7143
FSRCNN [30]	36.94 0.9551	33.13 0.9135	30.69 0.8656	**32.57** **0.9081**	29.39 0.8237	27.56 0.7532	**31.48** **0.8899**	28.50 **0.7888**	26.95 0.7126
SML-SR-NLM	36.86 0.9499	33.15 0.9110	30.97 0.8746	32.30 0.9007	29.33 0.8186	27.58 0.7561	31.24 0.8800	28.41 0.7789	26.96 0.7117
SML-SR-Baseline	36.76 0.9550	33.22 0.9147	31.07 0.8745	32.45 0.9062	29.48 0.8236	27.68 0.7566	31.29 0.8872	28.44 0.7866	26.99 0.7138
SML-SR	**37.17** **0.9558**	**33.50** **0.9175**	**31.26** **0.8791**	32.56 0.9076	**29.58** **0.8262**	**27.76** **0.7593**	31.38 0.8883	**28.55** 0.7887	**27.06** **0.7155**

interpolation function (implemented with the imresize function in matlab) with scaling factors $1/s$ ($s = 2, 3$, or 4) to the original HR images. The bicubic interpolation function downsamples the image by generating the LR pixels as a weighted average of pixels in a 4×4 neighborhood. This type of bicubic down-sampling has been widely adopted in recent literature of image SR [20], [22], [25], [29], [40]. In our implementation, to deal with this type of down-sampling, the degradation operation $\boldsymbol{y} = \mathbf{H}\boldsymbol{x}$ is implemented by applying the bicubic interpolation function with scaling factor $1/s$ to $\boldsymbol{x}$, and the corresponding upscaling operation $\mathbf{H}^\top \boldsymbol{y}$ is implemented by applying the bicubic interpolation function with scaling factor s to $\boldsymbol{y}$. The average PSNR and SSIM results of the reconstructed HR images are shown in Table II. From Table II it can be seen that SRCNN [25] outperforms A+ [22] and SCSR [20]. The NCSR method exploiting nonlocal structural sparsity performs comparable with the SRCNN method. The FSRCNN method containing more layers performs much better than the SRCNN method. The proposed SML-SR-Baseline and SML-SR methods outperform the SML-SR-NLM method, demonstrating the effectiveness of learning expectations from external training images. By jointly estimating the variances and sparse codes, the proposed SML-SR method can achieve better results than SML-SR-Baseline method. The average PSNR gains of the SML-SR method over SRCNN method can be up to 0.77dB. Compared with FSRCNN, the proposed SML-SR method performs better for most cases and the average PSNR gains over FSRCNN can be up to 0.57dB. Notice that we use the FSRCNN models released by the authors, which were trained with using both the 91 training images used in [20] and the General-100 image set. Parts of the reconstructed HR images by the test methods are shown in Figs. 3 and 4, from which we can see that the HR images reconstructed by the proposed method are most visually pleasant than those of other competing methods.

Considering that images are often corrupted by noise, we have also tested the robustness of the competing methods for noisy LR images. The Gaussian noise of standard deviation of 5 is added to the simulated LR images. Since other competing methods, i.e., KK [40], A+ [22], SRCNN [25], and FSRCNN [29] cannot deal with noise, we first apply a denoising algorithm to the LR images to remove the noise before applying these SR methods. Specifically, the well-known BM3D denoising method [44], which is among the state-of-the-art denoising methods, is adopted. The SR results for the noisy LR images are reported in Table III. It can be

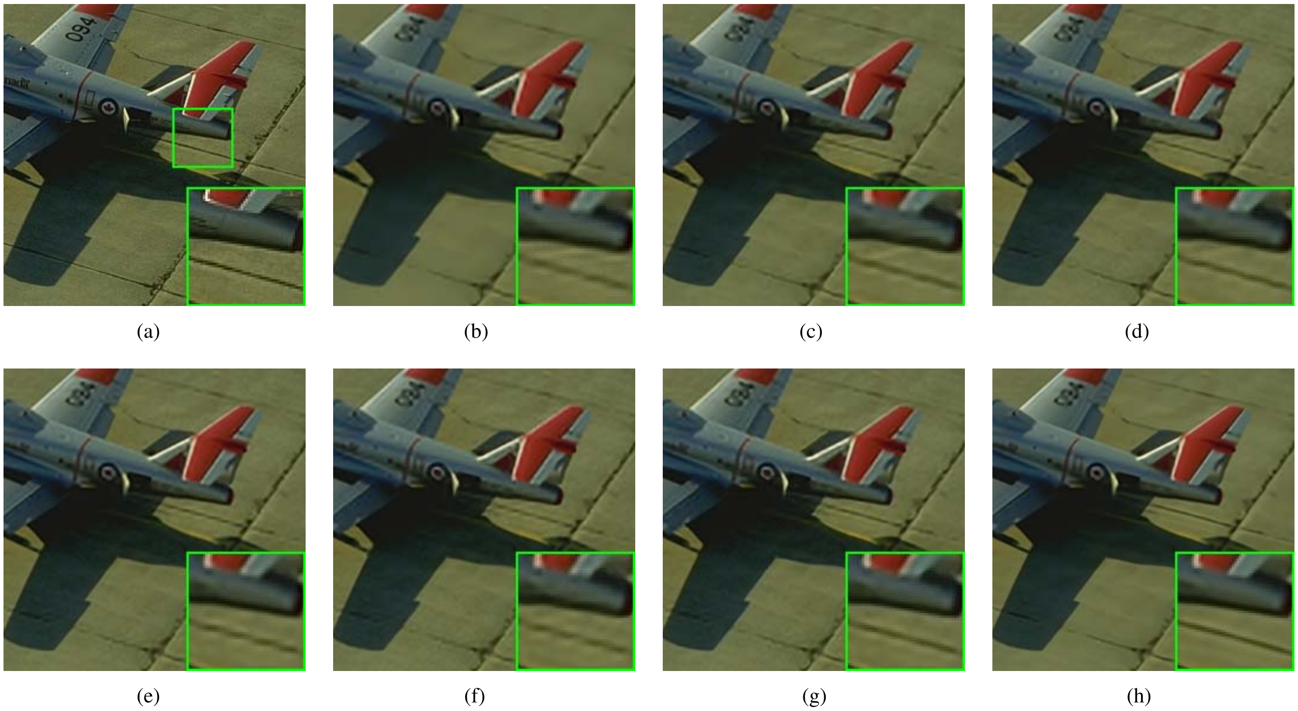


Fig. 3. SR results on image '37073' of BSD100 of scaling factor 3 (Bicubic downsampling and $\sigma_n = 5$). (a) Original. (b) SCSR [20] / 26.01dB. (c) KK [40] / 26.49dB. (d) A+ [22] / 30.77dB. (e) NCSR [13] / 30.94dB. (f) SRCNN [25] / 30.63dB. (g) FSRCNN [29] / 30.93dB. (h) Proposed SML-SR / 31.41dB.

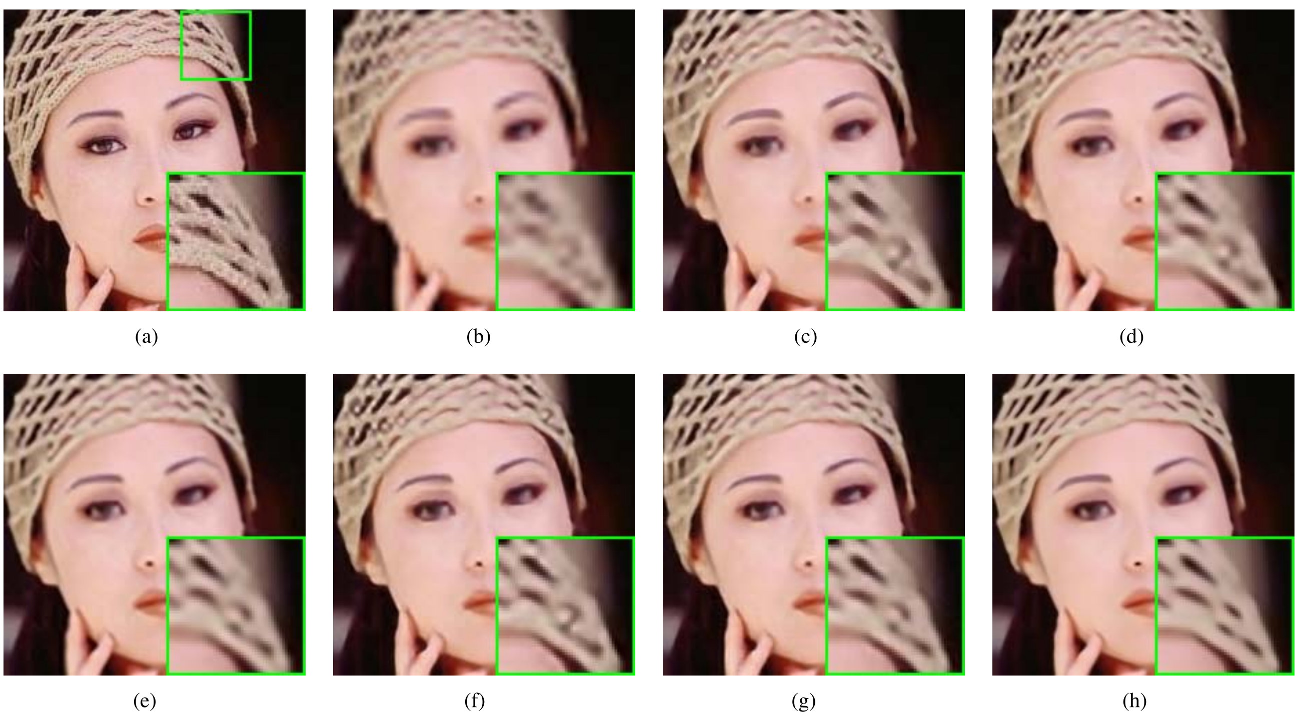


Fig. 4. SR results on image 'Woman' of Set5 of scaling factor 4 (Bicubic downsampling and $\sigma_n = 0$). (a) Original. (b) Bicubic. (c) KK [40] / 26.49dB. (d) A+ [22] / 28.65dB. (e) NCSR [13] / 29.33dB. (f) SRCNN [25] / 28.89dB. (g) FSRCNN [29] / 29.34dB. (h) Proposed SML-SR/ 30.31dB.

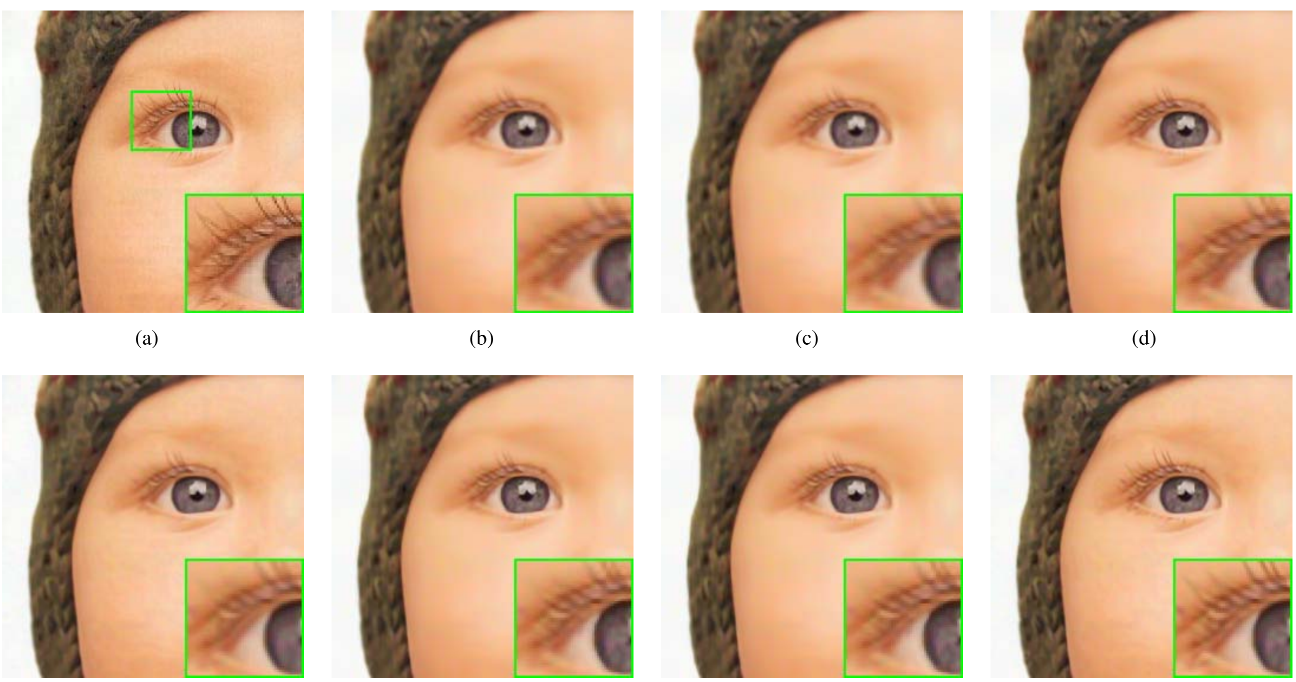
seen that the proposed method performs significantly better than other competing methods. The average PSNR gains of the proposed SML-SR method over FSRCNN method that is the second best method can be up to 0.77 dB. This demonstrates the advantages of performing super-resolution and denoising simultaneously. Parts of the reconstructed results are shown in Fig. 5. Obviously, edges and textures in reconstructed HR images by the proposed method are much sharper than those by other methods.

B. LR Images Generated With Gaussian Blur Followed by Downsampling

Another commonly used image downsampling method is Gaussian blurring followed by sub-sampling. In this paper,

TABLE III
AVERAGE PSNR AND SSIM RESULTS OF THE TEST METHODS (BICUBIC DOWNSAMPLING AND $\sigma_n = 5$)

Images	Se5			Set14			BSD100		
Upscaling	×2	×3	×4	×2	×3	×4	×2	×3	×4
SCSR [21]	-	30.42 0.8506	-	-	27.72 0.7597	-	-	27.19 0.7172	-
KK [41]	32.31 0.8953	29.56 0.8381	27.81 0.7853	29.44 0.8288	27.05 0.7410	25.63 0.6767	28.79 0.7951	26.72 0.6999	25.58 0.6385
A+ [23]	34.23 0.9151	31.30 0.8724	29.34 0.8275	30.97 0.8588	28.38 0.7787	26.77 0.7161	30.00 0.8295	27.62 0.7359	26.32 0.6714
SRCNN [26]	34.39 0.9153	31.47 0.8734	29.60 0.8322	31.16 0.8612	28.53 0.7810	26.95 0.7186	30.17 0.8329	27.73 0.7383	26.40 0.6729
NCSR [14]	34.24 0.9186	31.83 0.8848	29.93 0.8448	30.92 0.8583	28.64 0.7883	27.06 0.7279	29.90 0.8262	27.80 0.7462	26.52 0.6834
FSRCNN [30]	34.54 0.9161	31.74 0.8766	29.71 0.8333	31.25 0.8626	28.62 0.7828	27.00 0.7197	30.27 0.8352	27.80 0.7404	26.44 0.6742
SML-SR-NLM	34.64 0.9226	31.87 0.8852	29.96 0.8525	31.09 0.8718	28.61 0.7848	26.99 0.7314	30.25 0.8485	27.77 0.7397	26.52 0.6778
SML-SR-Baseline	34.82 0.9283	31.99 0.8885	30.27 0.8471	31.31 0.8748	28.71 0.7946	27.18 0.7314	30.31 0.8536	27.85 0.7556	26.57 0.6882
SML-SR	**35.05** **0.9304**	**32.23** **0.8923**	**30.48** **0.8572**	**31.38** **0.8769**	**28.87** **0.7980**	**27.30** **0.7371**	**30.43** **0.8555**	**27.99** **0.7579**	**26.74** **0.6913**



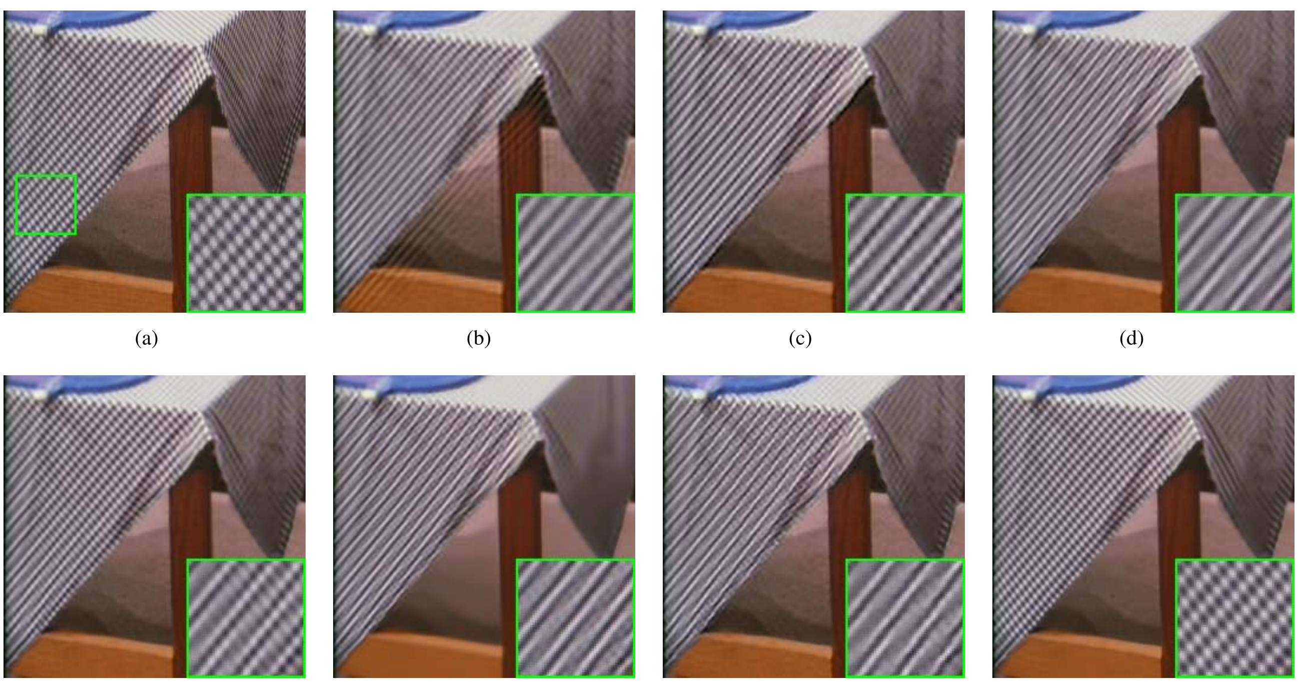
(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 5. SR results on image 'baby' of Set5 of scaling factor 2 (Bicubic downsampling and $\sigma_n = 5$). (a) Original. (b) Bicubic. (c) KK [40] / 35.1dB. (d) A+ [22] / 35.89dB. (e) NCSR [13] / 35.84dB. (f) SRCNN [25] / 35.92dB. (g) FSRCNN [29] / 35.92dB. (h) Proposed SML-SR / 36.53dB.

the 7×7 Gaussian kernel of standard deviation of 1.6 is used and the downsampling is performed with scaling factors $s = 2, 3, 4$ along both horizontal and vertical directions. For the SCSR and KK methods that cannot deal with the Gaussian blur kernel, we apply the iterative back-projection [45] method to the reconstructed HR images as a post processing step to remove the blur. For other learning-based SR methods (i.e., the A+, SRCNN and FSRCNN methods), for fair comparisons we have retrained their models using the LR images generated by Gaussian blurring and downsampling. The PSNR and SSIM

TABLE IV
AVERAGE PSNR AND SSIM RESULTS OF THE TEST METHODS (GAUSSIAN DOWNSAMPLING AND $\sigma_n = 0$)

Images	Se5			Set14			BSD100		
Upscaling	×2	×3	×4	×2	×3	×4	×2	×3	×4
SCSR [21]	-	30.22 0.8484	-	-	27.51 0.7619	-	-	27.10 0.7338	-
KK [41]	32.02 0.9045	30.28 0.8536	27.14 0.7802	29.36 0.8640	27.46 0.7640	25.00 0.6715	28.85 0.8188	27.10 0.7342	25.11 0.6439
A+ [23]	33.25 0.9214	32.13 0.9003	30.02 0.8579	29.91 0.8477	28.76 0.8067	27.19 0.7465	29.09 0.8163	28.04 0.7702	26.70 0.7066
SRCNN [26]	33.69 0.9252	32.29 0.8993	30.01 0.8559	30.14 0.8645	28.92 0.8098	27.20 0.7448	29.48 0.8428	28.14 0.7754	26.70 0.7058
NCSR [14]	35.15 0.9349	33.07 0.9121	30.53 0.8705	31.36 0.8756	29.30 0.8204	27.42 0.7543	30.24 0.8484	28.37 0.7839	26.79 0.7122
FSRCNN [30]	33.77 0.9281	32.77 0.9071	30.16 0.8583	30.76 0.8834	28.95 0.8135	27.15 0.7453	30.46 0.8744	28.26 0.7809	26.72 0.7076
SML-SR-NLM	35.14 0.9361	33.16 0.9120	30.56 0.8684	31.30 0.8805	29.30 0.8189	27.37 0.7486	30.32 0.8545	28.40 0.7795	26.77 0.7024
SML-SR-Baseline	35.29 0.9361	33.36 0.9135	30.76 0.8713	31.60 0.8797	29.55 0.8220	27.60 0.7532	30.43 0.8546	28.47 0.7856	26.82 0.7111
SML-SR	**35.55** **0.9389**	**33.58** **0.9173**	**30.97** **0.8777**	**31.68** **0.8826**	**29.61** **0.8254**	**27.66** **0.7579**	**30.54** **0.8572**	**28.57** **0.7882**	**26.93** **0.7141**



(a) (b) (c) (d)

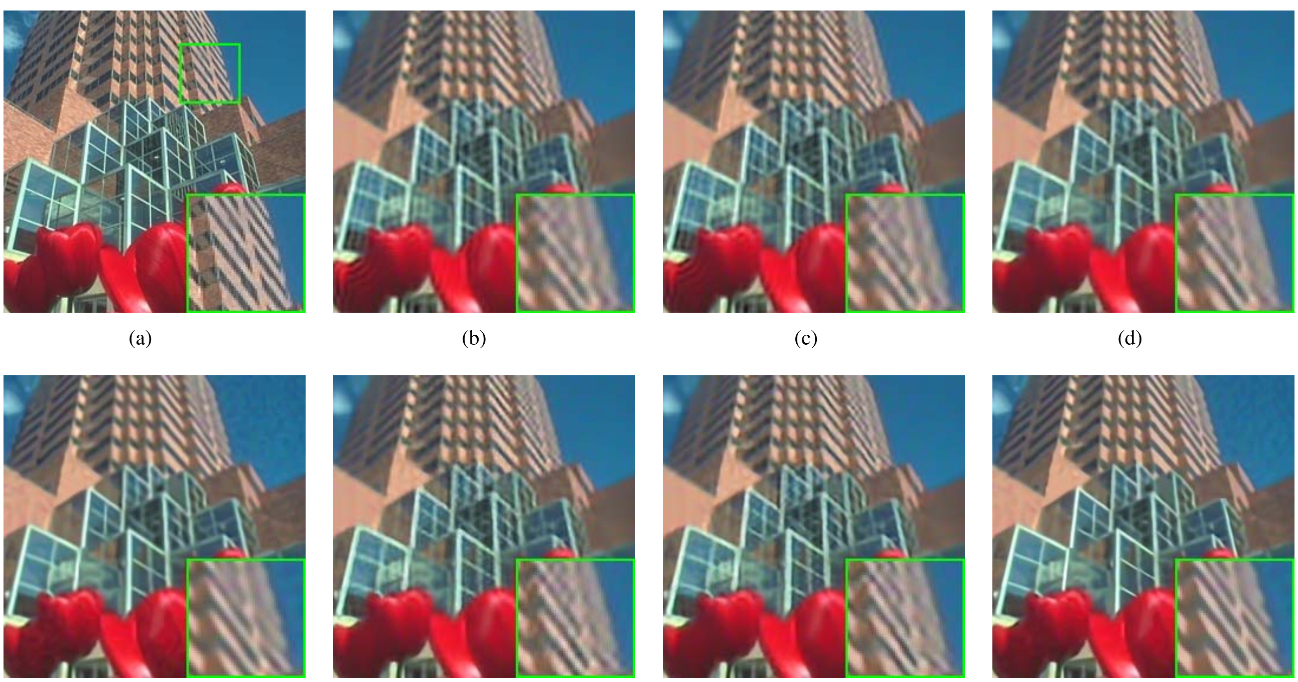
(e) (f) (g) (h)

Fig. 6. SR results on image 'Barbara' of Set14 of scaling factor 3 (Gaussian downsampling and $\sigma_n = 0$). (a) Original. (b) SCSR [20] / 26.01dB. (c) KK [40] / 26.49dB. (d) A+ [22] / 26.52dB. (e) NCSR [13] / 27.09dB. (f) SRCNN [25] / 26.66dB. (g) FSRCNN [29] / 26.57dB. (h) Proposed SML-SR / 27.31dB.

results are reported in Table IV. From Table IV, we can see that the learning-based methods, i.e., A+, SRCNN and FSRCNN methods perform much worse than the sparsity-based NCSR method. The SML-SR-Baseline and SML-SR methods outperform the SML-SR-NLM method, showing the effectiveness of learning expectations from training HR images. By jointly estimating the variances and the sparse codes, the proposed SML-SR method can further improve the SR performance over SML-SR-Baseline method. The average PSNR gains of the proposed SML-SR method over the NCSR method can be up to 0.51 dB. Parts of the reconstructed HR images are shown in Fig.6, from which we can see that the proposed method can reconstruct finer details and shaper edges than all the other competing methods.

TABLE V
AVERAGE PSNR AND SSIM RESULTS OF THE TEST METHODS (GAUSSIAN DOWNSAMPLING AND $\sigma_n = 5$)

Images	Se5			Set14			BSD100		
Upscaling	×2	×3	×4	×2	×3	×4	×2	×3	×4
SCSR [21]	-	29.27 0.8169	-	-	26.85 0.7188	-	-	26.51 0.6803	-
KK [41]	30.84 0.8626	29.33 0.8241	27.18 0.7601	28.15 0.7735	26.85 0.7228	24.93 0.6512	27.55 0.7326	26.52 0.6837	25.10 0.6232
A+ [23]	31.20 0.8726	30.37 0.8504	28.99 0.8211	28.44 0.7782	27.66 0.7471	26.58 0.70.83	27.68 0.7338	27.01 0.7004	26.15 0.6637
SRCNN [26]	30.89 0.8575	30.59 0.8516	29.12 0.8223	28.01 0.7648	27.83 0.7494	26.65 0.70.85	27.59 0.7343	27.11 0.70.37	26.18 0.6640
NCSR [14]	32.11 0.8840	31.11 0.8663	29.68 0.8397	29.05 0.7960	28.11 0.7635	26.91 0.7216	28.17 0.7568	27.37 0.7208	26.37 0.6770
FSRCNN [30]	31.20 0.8638	30.69 0.8536	29.13 0.8220	28.36 0.7735	27.91 0.7525	26.57 0.7075	27.79 0.7382	27.20 0.7068	26.18 0.6645
SML-SR-NLM	32.09 0.8869	31.07 0.8689	29.84 0.8455	28.95 0.7957	28.06 0.7649	26.94 0.7250	28.31 0.7554	27.38 0.7207	26.41 0.6790
SML-SR-Baseline	32.32 0.8851	31.35 0.8659	29.94 0.8453	29.14 0.7941	28.26 0.7644	27.07 0.7266	28.19 0.7544	27.41 0.7235	26.44 0.6835
SML-SR	**32.49 0.8896**	**31.55 0.8734**	**30.07 0.8500**	**29.30 0.7982**	**28.42 0.7708**	**27.13 0.7302**	**28.30 0.7575**	**27.57 0.7284**	**26.51 0.6859**



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 7. SR results on image '86000' of BSD100 of scaling factor 3 (Gaussian downsampling and $\sigma_n = 5$). (a) Original. (b) SCSR [20] / 25.18dB. (c) KK [40] / 25.21dB. (d) A+ [22] / 25.65dB. (e) NCSR [13] / 26.18dB. (f) SRCNN [25] / 25.69dB. (g) FSRCNN [29] / 25.73dB. (h) Proposed SML-SR / 26.72dB.

To test the robustness of different competing methods, additive Gaussian noise of standard deviation of 5 is added to the simulated LR images. Similarly, the BM3D denoising method is applied to the LR images before applying the KK, A+, SRCNN and FSRCNN methods. The SR results are reported in Table V, from which we can see that the NCSR method performs much better than other competing methods. The average PSNR gains of the proposed SML-SR method over the NCSR method can be up to 0.44 dB. Parts of the reconstructed HR images are shown in Fig. 7. It is obviously that the HR image reconstructed by the proposed method is much clear and shaper than other methods.

Regarding the computational complexity, the proposed method usually requires 100 iterations to converge. Therefore, the computational complexity is higher than those non-iterative methods such as A+, SRCNN and FSRCNN methods. For a 256×256 test image, the running time of the proposed method is about 96 seconds (Matlab implementation on an i7 3.4GHz CPU). Since the patch grouping and sparse coding for each group can be implemented in parallel, the proposed method can be accelerated by using parallel computing techniques on a GPU platform.

V. Conclusion

In this paper, a novel hybrid parametric sparse model learning method is proposed for image super-resolution. Specifically, a set of mapping functions between the LR patches and the sparse codes of the HR patches are learned from a training set. Then, the parametric sparse distributions can be estimated from both the learned sparse codes and those estimated from the LR image. With the learned sparse models, the sparse codes of the HR image patches and thus the HR image can then be faithfully recovered from the LR observation. Experimental results show that the proposed SR method is competitive with and often better than the current state-of-the-art image SR methods. Also, the proposed method can perform image SR and denoising simultaneously, showing the robustness to additive noise.

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Yongbo Li received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 2012, where he is currently pursuing the Ph.D. degree with the School of Artificial Intelligence.

His research interests include image restoration and sparse representation.

Weisheng Dong (M'11) received the B.S. degree in electronics engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in circuits and system from Xidian University, Xi'an, China, in 2010.

He was a Visiting Student with Microsoft Research Asia, Beijing, China, in 2006. From 2009 to 2010, he was a Research Assistant with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong. In 2010, he joined the School of Electronic Engineering, Xidian University, as a Lecturer, where he has been an Associate Professor since 2012. His current research interests include inverse problems in image processing, sparse signal representation, and image compression.

Dr. Dong was a recipient of the Best Paper Award at the SPIE Visual Communication and Image Processing in 2010. He is currently serving as an Associate Editor of the IEEE TRANSACTIONS ON IMAGE PROCESSING.

Xuemei Xie received the M.S. degree in electronic engineering from Xidian University in 1994 and the Ph.D. degree in electrical and electronic engineering from The University of Hong Kong in 2004.

Her research interests are digital signal processing, multi-rate filter banks, and wavelet transforms.

Guangming Shi (SM'10) received the B.S. degree in automatic control, the M.S. degree in computer control, and the Ph.D. degree in electronic information technology from Xidian University in 1985, 1988, and 2002, respectively.

He joined the School of Electronic Engineering, Xidian University, in 1988. From 1994 to 1996, as a Research Assistant, he cooperated with the Department of Electronic Engineering, The University of Hong Kong. Since 2003, he has been a Professor with the School of Electronic Engineering, Xidian University, and the Head of National Instruction Base of Electrician & Electronic in 2004. In 2004, he had studied in the Department of Electronic Engineering, University of Illinois at Urbana–Champaign. He is currently the Deputy Director of the School of Electronic Engineering, Xidian University, and the academic leader in the subject of circuits and systems. He has authored or co-authored over 60 research papers. His research interests include compressed sensing, theory and design of multirate filter banks, image denoising, low-bit-rate image/video coding and implementation of algorithms for intelligent signal processing (using DSP and FPGA).

Jinjian Wu received the B.Sc. and Ph.D. degrees from Xidian University, Xi'an, China, in 2008 and 2013, respectively.

From 2011 to 2013, he was a Research Assistant with Nanyang Technological University, Singapore. From 2013 to 2014, he was a Post-Doctoral Research Fellow with Nanyang Technological University. From 2013 to 2015, he was a Lecturer with Xidian University. Since 2015, he has been an Associate Professor with the School of Electronic Engineering, Xidian University.

His research interests include visual perceptual modeling, saliency estimation, quality evaluation, and just noticeable difference estimation. He has served as the TPC Member of ICME2014, ICME2015, PCM2015, and ICIP2015. He received the Best Student Paper of ISCAS 2013.

Xin Li received the B.S. degree (Hons.) in electronic engineering and information science from the University of Science and Technology of China, Hefei, China, in 1996, and the Ph.D. degree in electrical engineering from Princeton University, Princeton, NJ, USA, in 2000.

He was a Technical Staff Member with Sharp Laboratories of America, Camas, WA, USA, from 2000 to 2002. Since 2003, he has been a Faculty Member with the Lane Department of Computer Science and Electrical Engineering. His research interests include image/video coding and processing. He was a recipient of the Best Student Paper Award at the Conference of Visual Communications and Image Processing in 2001, the Best Student Paper Award at the IEEE Asilomar Conference on Signals, Systems and Computers in 2006, and the Best Paper Award at the Conference of Visual Communications and Image Processing in 2010. He is currently serving as a member of the Image, Video and Multidimensional Signal Processing Technical Committee and an Associate Editor of the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY.